



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2007

**TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION**

Mathematics Extension 1

General Instructions

- Reading Time – 5 Minutes
- Working time – 2 Hours
- Write using black or blue pen. Pencil may be used for diagrams.
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown in every question.
- Each Question is to be returned in a separate bundle.

Total Marks – 84

- Attempt Questions 1 – 7.
- All questions are of equal value.

Examiner: *A. Fuller*

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Total marks - 84

Attempt Questions 1-7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Marks

Question 1 (12 marks) Use a SEPARATE writing booklet.

- (a) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 4x}{5x}$. 1
- (b) Calculate the acute angle (to the nearest minute) between the lines $2x + y = 4$ and $x - 3y = 6$. 2
- (c) (i) Show that $x + 1$ is a factor of $x^3 - 4x^2 + x + 6$. 1
- (ii) Hence, or otherwise factorise $x^3 - 4x^2 + x + 6$ fully. 2
- (d) The point $P(5, 7)$ divides the interval joining the points $A(-1, 1)$ and $B(3, 5)$ externally in the ratio $k : 1$. Find the value of k . 2
- (e) Find the horizontal asymptote of the function $y = \frac{3x^2 - 4x + 1}{2x^2 - 1}$. 1
- (f) Find a primitive of $\frac{1}{\sqrt{4 - x^2}}$. 1
- (g) Solve the equation $|x + 1|^2 - 4|x + 1| - 5 = 0$. 2

Question 2 (12 marks)

- (a) Let $f(x) = \frac{1}{2} \cos^{-1}\left(\frac{x}{3}\right)$.
- (i) State the domain and range of the function $f(x)$. 2
- (ii) Show that $y = f(x)$ is a decreasing function. 2
- (iii) Find the equation of the tangent to the curve $y = f(x)$ at the point where $x = 0$. 2
- (b) Find the derivative of $y = \ln(\sin^3 x)$. 2
- (c) (i) Write $\cos x - \sqrt{3} \sin x$ in the form $A \cos(x + \alpha)$, where $A > 0$ and $0 < \alpha < \frac{\pi}{2}$. 2
- (ii) Hence, or otherwise, solve $\cos x - \sqrt{3} \sin x + 1 = 0$ for $0 \leq x \leq 2\pi$. 2

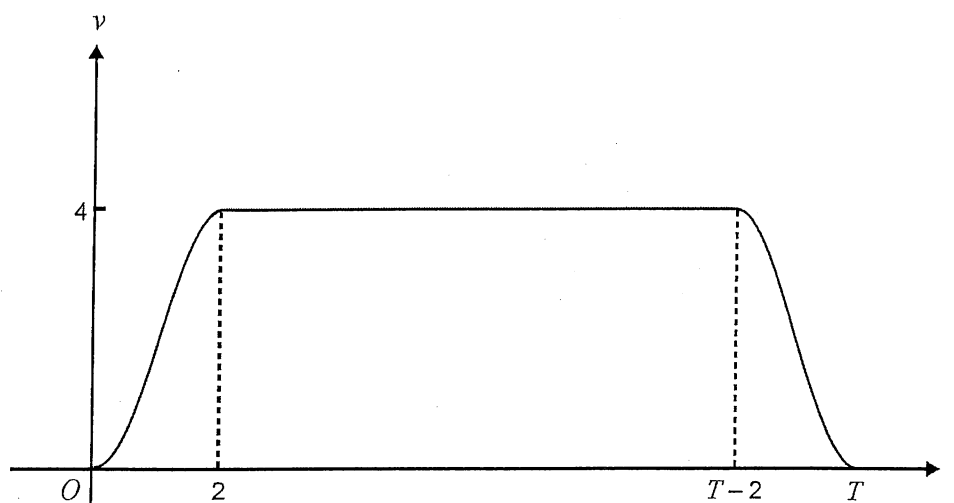
Question 3 (12 marks) Use a SEPARATE writing booklet.

- | | | | |
|-----|---|---|---|
| (a) | (i) | Show that the equation $e^x - x - 2 = 0$ has a solution in the interval $1 < x < 2$. | 1 |
| | (ii) | Taking an initial approximation of $x = 1.5$ use one application of Newton's method to approximate the solution, correct to three decimal places. | 2 |
| (b) | The normal at $P(2ap, ap^2)$ on the parabola $x^2 = 4ay$ cuts the y -axis at Q and is produced to a point R such that $PQ = QR$. | | |
| | (i) | Show that the equation of the normal at P is $x + py = 2ap + ap^3$. | 2 |
| | (ii) | Find the coordinates of Q . | 1 |
| | (iii) | Show that R has coordinates $(-2ap, ap^2 + 4a)$. | 1 |
| | (iv) | Show that the locus of R is a parabola, and find its vertex. | 3 |
| (c) | If $\int_1^5 f(x)dx = 3$, find $\int_1^5 (2f(x) + 1)dx$. | | 2 |

Question 4 (12 marks) Use a SEPARATE writing booklet.

(a) Using the substitution $u = e^x$, or otherwise, find $\int e^{(e^x+x)} dx$ 3

- (b) The velocity-time graph below shows the velocity of a lift as it travels from the first floor to the twentieth floor of a tall building during the T seconds of its motion.



The velocity v m/s at time t s for $0 \leq t \leq 2$ is given by $v = t^2(3-t)$. After the first two seconds, the lift moves with a constant velocity of 4 m/s for a time, and then decelerates to rest in the final two seconds.

The velocity-time graph is symmetrical about $t = \frac{1}{2}T$.

- (i) Express the acceleration in terms of t for the first two seconds of the motion of the lift. 1
- (ii) Hence, find the maximum acceleration of the lift during the first two seconds of its motion. 2
- (iii) Given that the total distance travelled by the lift during its journey is 41 metres, find the exact value of T . 2

- (c) A solid is formed by rotating about the y -axis the region bounded by the curve $y = \cos^{-1} x$, the x -axis and the y -axis.

(i) Show that the volume of the solid is given by $V = \pi \int_0^{\frac{\pi}{2}} \cos^2 y dy$. **1**

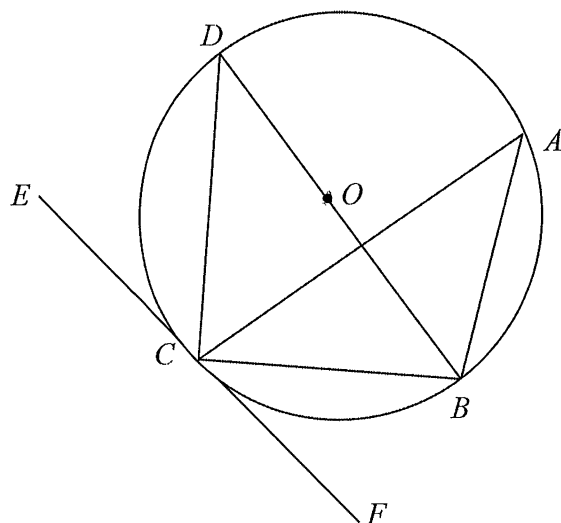
(ii) Calculate the volume of this solid. **3**

Question 5 (12 marks) Use a SEPARATE writing booklet.

- (a) Use mathematical induction to prove that $\sum_{r=1}^n r \times r! = (n+1)! - 1$. 3
- (b) In the expansion of $\left(2x + \frac{1}{x^2}\right)^{15}$, determine the coefficient of the term that is independent of x . 3
- (c) The acceleration of a particle P is given by the equation $a = 8x(x^2 + 1)$, where x is the displacement of P from the origin in metres after t seconds, with movement being in a straight line. Initially, the particle is projected from the origin with a velocity of 2 metres per second in the negative direction.
- (i) Show that the velocity of the particle can be expressed as $v = 2(x^2 + 1)$. 2
- (ii) Hence, show that the equation describing the displacement of the particle at time t is given by $x = \tan 2t$. 2
- (iii) Determine the velocity of the particle after $\frac{\pi}{8}$ seconds. 2

Question 6 (12 marks) Use a SEPARATE writing booklet.

(a)



A, B, C and D are points on the circumference of a circle with centre O .
 EF is a tangent to the circle at C and the angle ECD is 60° .

3

Find the value of $\angle BAC$ giving reasons.

- (b) (i) By considering the expansion of $(1+x)^n$ in ascending powers of x , where n is a positive integer, and differentiating, show that

1

$$\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \dots + n\binom{n}{n} = n(2^{n-1}).$$

- (ii) Hence, find an expression for $2\binom{n}{1} + 3\binom{n}{2} + 4\binom{n}{3} + \dots + (n+1)\binom{n}{n}$.

2

- (c) If $f(x+2) = x^2 + 2$, find $f(x)$.

2

- (d) At a particular dinner, each rectangular table has nine seats, five facing the stage and four with their backs to the stage.

In how many ways can 9 people be seated at the table if

- (i) John and Mary sit on the same side?

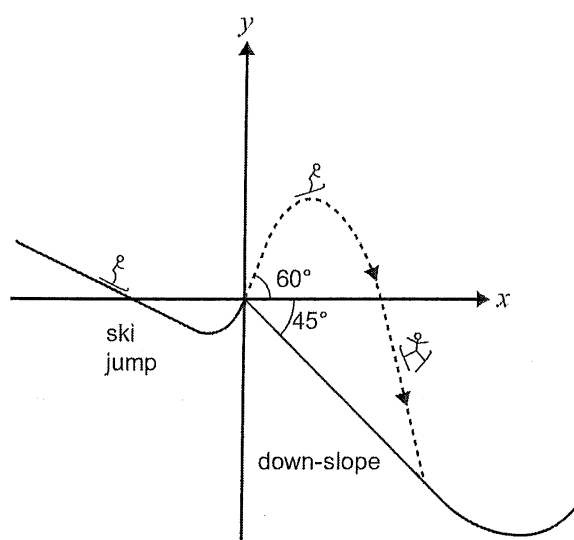
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- (ii) John and Mary sit on opposite sides?

2

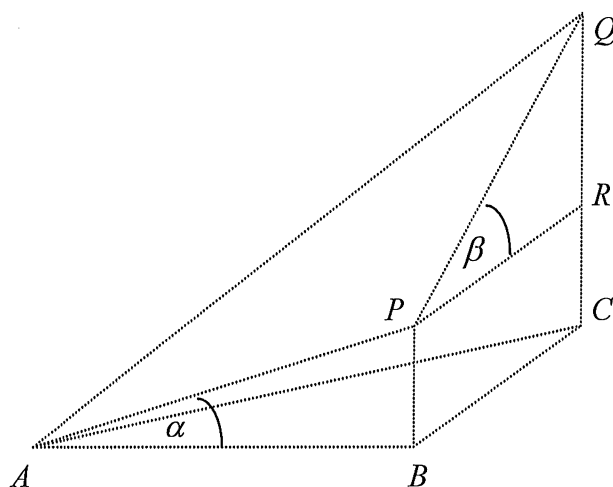
Question 7 (12 marks) Use a SEPARATE writing booklet.

- (a) A skier accelerates down a slope and then skis up a short jump (see diagram). The skier leaves the jump at a speed of 12 m/s and at an angle of 60° to the horizontal. The skier performs various gymnastic twists and lands on a straight line section of the 45° down-slope T seconds after leaving the jump. Let the origin O of a Cartesian coordinate system be at the point where the skier leaves the jump. Displacements are measured in metres and time in seconds. Let $g = 10 \text{ m s}^{-2}$ and neglect air resistance.



- | | | |
|-------|---|---|
| (i) | Derive the cartesian equation of the skiers flight as a function of y in terms of x . | 3 |
| (ii) | Show that $T = \frac{6}{5}(\sqrt{3} + 1)$. | 3 |
| (iii) | At what speed, in metres per second does the skier land on the down-slope? Give your answer correct to one decimal place. | 2 |

(b)



ABC is a horizontal, right-angled, isosceles triangle where $AB = BC$ and $\angle ABC = 90^\circ$. P is vertically above B ; Q is vertically above C . The angle of elevation of P from A , and Q from P are α and β respectively.

- (i) If the angle of elevation of Q from A is θ , prove that

2

$$\tan \theta = \frac{\tan \alpha + \tan \beta}{\sqrt{2}}.$$

- (ii) If $\angle APQ = \phi$, prove that $\cos \phi = -\sin \alpha \sin \beta$.

2

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

2007 THSC Mathematics Extension 1: Solutions— Question 1

1. (a) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 4x}{5x}$.

1

Solution:

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sin 4x}{5x} &= \lim_{x \rightarrow 0} \frac{\sin 4x}{4x} \times \frac{4}{5}, \\ &= \frac{4}{5} \times \lim_{x \rightarrow 0} \frac{\sin 4x}{4x}, \\ &= \frac{4}{5}.\end{aligned}$$

- (b) Calculate the acute angle (to the nearest minute) between the lines $2x + y = 4$ and $x - 3y = 6$.

2

Solution:

$$\begin{aligned}\tan \alpha &= \frac{|-2 - 1/3|}{1 + (-2) \times (1/3)}, \\ &= 7, \\ \therefore \alpha &= \tan^{-1} 7, \\ &= 81.86989765^\circ \text{ by calculator,} \\ &= 81^\circ 52' .\end{aligned}$$

- (c) i. Show that $x + 1$ is a factor of $x^3 - 4x^2 + x + 6$.

1

Solution: Putting $P(x) = x^3 - 4x^2 + x + 6$;

$$\begin{aligned}P(-1) &= -1 - 4 - 1 + 6, \\ &= 0. \\ \therefore x + 1 &\text{ is a factor.}\end{aligned}$$

- ii. Hence or otherwise factorise $x^3 - 4x^2 + x + 6$ fully.

2

Solution: Possible factors of 6 are 1, 2, 3 or 1, -2, -3.

$$\begin{aligned}P(-2) &= -8 - 16 - 2 + 6 \neq 0, \\ P(2) &= 8 - 16 + 2 + 6, \\ &= 0. \\ \therefore x^3 - 4x^2 + x + 6 &= (x + 1)(x - 2)(x - 3).\end{aligned}$$

- (d) The point $P(5, 7)$ divides the interval joining the points $A(-1, 1)$ and $B(3, 5)$ externally in the ratio $k : 1$. Find the value of k .

2

Solution:
$$\frac{5 - (-1)}{5 - 3} = \frac{k}{1},$$
$$6 = 2k,$$
$$k = 3.$$

- (e) Find the horizontal asymptote of the function $y = \frac{3x^2 - 4x + 1}{2x^2 - 1}$.

1

Solution:
$$\lim_{x \rightarrow \pm\infty} \frac{3 - 4/x + 1/x^2}{2 - 1/x^2} = \frac{3}{2}.$$
$$\therefore y = 3/2 \text{ is the horizontal asymptote.}$$

- (f) Find a primitive of $\frac{1}{\sqrt{4 - x^2}}$.

1

Solution: From the table of standard integrals,

$$\int \frac{dx}{\sqrt{4 - x^2}} = \sin^{-1} \frac{x}{2} + c.$$

- (g) Solve the equation $|x + 1|^2 - 4|x + 1| - 5 = 0$.

2

Solution: Putting $y = |x + 1|$;
 $y^2 - 4y - 5 = 0,$
 $(y - 5)(y + 1) = 0,$
 $\therefore y = 5 \text{ or } -1.$
But $|x + 1| \geq 0,$
hence $x + 1 = 5$ or $x + 1 = -5,$
so $x = 4, -6.$

(a) i) $f(x) = \frac{1}{2} \cos^{-1}\left(\frac{x}{3}\right)$

Range: $0 \leq \cos^{-1}x \leq \pi$

$y = \cos^{-1}x$ has $-1 \leq x \leq 1$ and $0 \leq y \leq \pi$

$\frac{1}{2} \times 0 \leq \frac{1}{2} \cos^{-1}\left(\frac{x}{3}\right) \leq \frac{1}{2} \times \pi$

Domain $-1 \leq x \leq 1$
 $-1 \leq \frac{x}{3} \leq 1$
 $-3 \leq x \leq 3$ ①

$0 \leq f(x) \leq \frac{\pi}{2}$ ①

ii) $f'(x) = \frac{1}{2} \times \frac{-1}{\sqrt{1 - \frac{x^2}{9}}} \times \frac{1}{3}$

$= -\frac{1}{6} \times \frac{1}{\sqrt{\frac{9-x^2}{9}}} = -\frac{1}{6} \times \frac{3}{\sqrt{9-x^2}} = -\frac{1}{2} \cdot \frac{1}{\sqrt{9-x^2}}$

So for $-3 < x < 3$, $f'(x) < 0$ always. ②

iii) when $x=0$, $f(x) = \frac{1}{2} \cos^{-1}\left(\frac{0}{3}\right)$
 $= \frac{1}{2} \cos^{-1}(0)$
 $= \frac{1}{2} \times \frac{\pi}{2} = \frac{\pi}{4}$

$f'(x) = \frac{-1}{2\sqrt{9-x^2}}$

at $x=0$, $m = \frac{-1}{2 \times 3} = -\frac{1}{6}$

$(y - y_1) = m(x - x_1)$

$(y - \frac{\pi}{4}) = -\frac{1}{6}(x - 0)$

$y = -\frac{1}{6}x + \frac{\pi}{4}$

or $\frac{1}{6}x + y - \frac{\pi}{4} = 0$

or $12 \times \frac{1}{6}x + 12y - 12 \times \frac{\pi}{4} = 0$

$2x + 12y - 3\pi = 0$ ②

(b) $y = \ln(\sin^3 x)$

$y' = \frac{1}{\sin^3 x} \times 3\sin^2 x \times \cos x \times 1$

$= \frac{3\sin^2 x \cos x}{\sin^3 x}$

$= 3 \frac{\cos x}{\sin x}$

$= 3 \cot x$ ②

(C) (i) $1 \cos x - \sqrt{3} \sin x$

$$A = \sqrt{1^2 + (-\sqrt{3})^2} = \sqrt{4} = 2$$

$$= 2 \left(\frac{1}{2} \cos x - \frac{\sqrt{3}}{2} \sin x \right)$$

$$= A (\cos x \cos \alpha - \sin x \sin \alpha)$$

$$\Rightarrow A = 2 \quad (1)$$

$$\left. \begin{array}{l} \cos \alpha = \frac{1}{2} \\ \sin \alpha = \frac{\sqrt{3}}{2} \end{array} \right\} \alpha = 60^\circ = \frac{\pi}{3} \quad (1) \quad (1^{\text{st}} \text{ quad } 0 < \alpha < \frac{\pi}{2})$$

$$\text{So } \cos x - \sqrt{3} \sin x = 2 \cos \left(x + \frac{\pi}{3} \right)$$

(ii) Now $\cos x - \sqrt{3} \sin x + 1 = 0$
 $\cos x - \sqrt{3} \sin x = -1$

$$2 \cos \left(x + \frac{\pi}{3} \right) = -1$$

$$\cos \left(x + \frac{\pi}{3} \right) = -\frac{1}{2}$$

$$\cos \frac{\pi}{3} = \frac{1}{2}$$

2nd / 3rd quad.

now $x + \frac{\pi}{3} = \pi - \frac{\pi}{3}$

2nd quad.

and $x + \frac{\pi}{3} = \pi + \frac{\pi}{3}$

3rd quad.

$$\text{So } x = \pi - \frac{2\pi}{3} = \frac{3\pi - 2\pi}{3} = \frac{\pi}{3} \quad \checkmark$$

$$\text{and } x = \pi + \frac{\pi}{3} - \frac{\pi}{3} = \pi \quad \checkmark \quad (2)$$

Q3 (a) (i) let $f(x) = e^x - x - 2$.

now $f(1) = e - 1 - 2 = e - 3 < 0$. (~ -0.28)

& $f(2) = e^2 - 2 - 2 = e^2 - 4 > 0$. (~ 3.38).

Since $f(x)$ changes sign in $1 < x < 2$ ($\checkmark$)

$f(x) = 0$ has a solution in $1 < x < 2$.

(ii) now $x_r = x_1 - \frac{f(x_1)}{f'(x_1)}$ & $f(x) = e^x - x - 2$
 $f'(x) = e^x - 1$.

$$\therefore x_r = 1.5 - \frac{f(1.5)}{f'(1.5)}$$

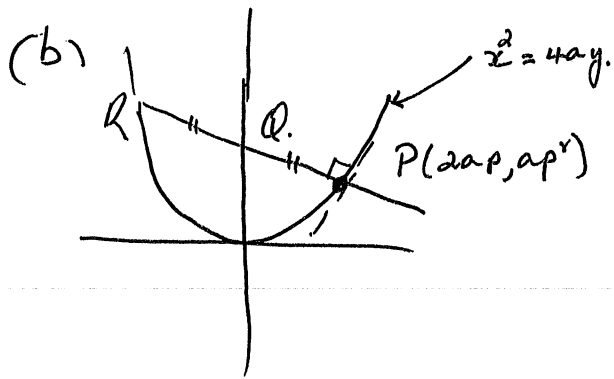
$$= 1.5 - \frac{(e^{1.5} - 1.5 - 2)}{e^{1.5} - 1}$$

$$\doteq 1.5 - \frac{0.98168}{3.481689}$$

$$\doteq 1.5 - 0.281915$$

$$\doteq \underline{1.218}$$

($\checkmark\checkmark$)



(i) $y = \frac{1}{4a} x^2$

$$y' = \frac{1}{2a} x$$

$$\therefore m_T = \frac{2ap}{2a} = p$$

$$\therefore m_N = -\frac{1}{p}$$

H: $\frac{y - ap^2}{x - 2ap} = -\frac{1}{p}$

$$py - ap^3 = -x + 2ap$$

$$\boxed{x + py = 2ap + ap^3} \quad (\checkmark \checkmark)$$

(ii) Co-ords of Q. $x = 0 \quad \therefore py = 2ap + ap^3$

$$y = 2a + ap^2$$

$$\therefore Q(0, 2a + ap^2)$$

(✓).

(iii) Q is the mid-pt of PR.

let R be (x_1, y_1)

$$\therefore \frac{x_1 + 2ap}{2} = 0$$

$$\therefore x_1 = -2ap$$

$$\frac{y_1 + ap^2}{2} = 2a + ap^2$$

$$y_1 + ap^2 = 4a + 2ap^2$$

$$y_1 = ap^2 + 4a$$

$$\therefore R(-2ap, ap^2 + 4a) \quad (\checkmark)$$

(iii) To find the locus of R. $x = -2ap \quad \therefore p = \frac{x}{-2a}$

$$\therefore y = a\left(\frac{x}{-2a}\right)^2 + 4a$$

$$y = \frac{x^2}{4a} + 4a$$

$$4ay = x^2 + 16a^2$$

$$x^2 = 4ay - 16a^2$$

$$x^2 = 4a(y - 4a)$$

PARABOLA VERTEX
(0, 4a).
(✓✓✓)

(c)

$$\int_1^5 (2f(x) + 1) dx = 2 \int_1^5 f(x) dx + \int_1^5 1 \cdot dx$$

$$= 2 \times 3 + [x]_1^5$$

$$= 6 + (5 - 1).$$

$$= 10.$$

—

(✓✓)

Question (f)

(a) Let $u = e^x$.

$$du = e^x dx$$

$$\int e^{(e^x + x)} dx$$

$$= \int e^{e^x} \cdot e^x dx$$

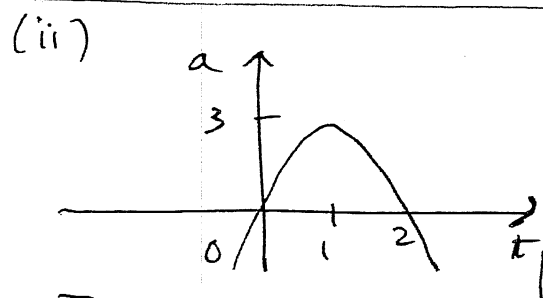
$$= \int e^u du = e^u + C$$

$$\Rightarrow e^{e^x} + C$$

(b) For $0 \leq t \leq 2$

(i) $v = 3t^2 - t^3$

$$a = \frac{dv}{dt} = \frac{6t - 3t^2}{3t(2-t)}$$



From the graph of a versus t .

The max. acceleration occurs when $t=1$

$$\therefore a_{\max} = 3 \times 1 = 3 \text{ m/s}^2$$

(iii) Let $d(t)$ be the total distance travelled (which is 41m).

$$\therefore d(t) = 2 \int_0^2 v(t) dt + \int_2^{T-2} 4 dt$$

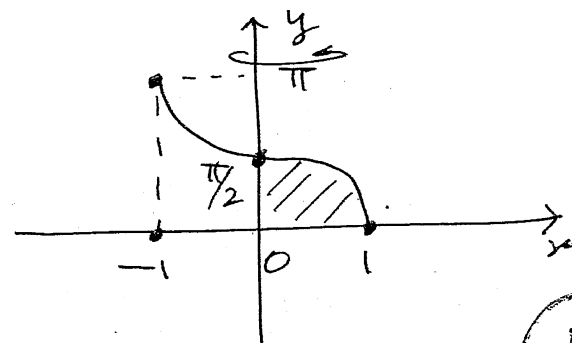
$$\therefore 41 = 2 \int_0^2 (3t^2 - t^3) dt + [4t]_2^{T-2}$$

$$\therefore 41 = 2 \left[t^3 - \frac{t^4}{4} \right]_0^2 + 4(T-2)$$

$$\therefore 41 = 2(8-4) + 4T - 8$$

$$41 = 4T - 8 \Rightarrow T = 12\frac{1}{4}$$

(c)



$$y = \cos^{-1}x$$

$$\Rightarrow x = \cos y, \begin{cases} -1 \leq x \leq 1 \\ 0 \leq y \leq \pi \end{cases}$$

$$\therefore V = \pi \int_0^{\pi/2} x^2 dy$$

$$\therefore x = \cos y, V = \pi \int_0^{\pi/2} \cos^2 y dy$$

(ii)

$$V = \pi \int_0^{\pi/2} \cos^2 y dy$$

$$\text{but } \cos^2 y = \left(\frac{1 + \cos 2y}{2} \right)$$

$$= \frac{\pi}{2} \int_0^{\pi/2} (1 + \cos 2y) dy$$

$$= \frac{\pi}{2} \left[y + \frac{\sin 2y}{2} \right]_0^{\pi/2}$$

$$= \frac{\pi}{2} \left(\frac{\pi}{2} \right) = \frac{\pi^2}{4}$$

EX1 QUESTION 5

(a) if $n=1$, $1 \times 1! = (1+1)! - 1$

$\therefore P(1)$ is true

Assume $P(k)$ is true $1 \times 1! + 2 \times 2! + \dots + k \times k! = (k+1)! - 1$

if $P(k+1)$ is $1 \times 1! + 2 \times 2! + \dots + k \times k! + (k+1)(k+1)! = (k+2)! - 1$

LHS is $(k+1)! - 1 + (k+1)(k+1)!$ using assumption
 $= (k+1)! (1 + k+1) - 1$
 $= (k+1)! (k+2) - 1 = (k+2)! - 1 = \text{RHS}$

$\therefore P(k+1)$ is true if $P(k)$ is true. $P(1)$ is true $\therefore$ by Mathematical Induction $\sum_{r=1}^n r \times r! = (n+1)! - 1$

(b)

$$T_{k+1} = {}^{15}C_k (2x)^{n-k} (x^{-2})^k$$

for term independent of x $n-k-2k=0$, $k=5$

$$\text{cf } {}^{15}C_5 \times 2^{10} = 3075072$$

(c) (i) $d\left(\frac{1}{2}v^2\right) = 8x(x^2+1) = 8x^3+8x$

$$\frac{1}{2}v^2 = 2x^4 + 4x^2 + C$$

$$v = -2 \quad x = 0 \quad C = 2$$

$$v^2 = 4x^4 + 8x^2 + 4 = 4(x^4 + 2x^2 + 1)$$

$$v = \pm 2(x^2+1)^2$$

(ii)

$$\text{if } \frac{dx}{dt} = 2(x^2+1)$$

$$\frac{dt}{dx} = \frac{1}{2(x^2+1)}$$

$$t = \frac{1}{2} \tan^{-1} x + C$$

$$t = 0 \quad x = 0, \quad C = 0$$

$$2t = \tan^{-1} x$$

$$x = \tan 2t$$

(iii)

$$t = \frac{\pi}{8}, \quad x = \tan \frac{\pi}{4} = 1$$

$$v = 2(1+1) \text{ from (i)}$$

$$v = 4 \text{ m/s}$$

Question 6

- (a) $\angle CBD = 60^\circ$ (alternate segment theorem)
 $\angle BCD = 90^\circ$ (angle in semicircle)
 $\therefore \angle CDB = 30^\circ$ (angle sum of triangle)
 $\therefore \angle CAB = 30^\circ$ (angles at circumference on same arc)

(b) (i) $(1+x)^n = 1 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_{n-1}x^{n-1} + x^n$

Differentiating with respect to x :

$$n(1+x)^{n-1} = {}^nC_1 + 2{}^nC_2x + 3{}^nC_3x^2 + \dots + n{}^nC_nx^{n-1}$$

Let $x = 1$:

$$n2^{n-1} = {}^nC_1 + 2{}^nC_2 + 3{}^nC_3 + \dots + n{}^nC_n$$

QED

(ii) Multiplying $(1+x)^n$ by x :

$$x(1+x)^n = {}^nC_0x + {}^nC_1x^2 + {}^nC_2x^3 + \dots + {}^nC_nx^{n+1}$$

Differentiating with respect to x :

$$xn(1+x)^{n-1} + (1+x)^n = {}^nC_0 + 2{}^nC_1x + 3{}^nC_2x^2 + \dots + (n+1){}^nC_nx^n$$

Let $x = 1$:

$$n(2)^{n-1} + (2)^n = 1 + 2{}^nC_1 + 3{}^nC_2 + \dots + (n+1){}^nC_n$$

Thus

$$\begin{aligned} 2{}^nC_1 + 3{}^nC_2 + \dots + (n+1){}^nC_n &= n(2)^{n-1} + (2)^n - 1 \\ &= (n+2)2^{n-1} - 1 \end{aligned}$$

(c) $f(x+2) = x^2 + 2$

$$f(x) = (x-2)^2 + 2$$

$$= x^2 - 4x + 4 + 2$$

$$= x^2 - 4x + 6$$

(d)



- (i) If J&M sit on the short side, they can be arranged in 12 ways, and the other guests in $7!$ ways. Thus $12 \times 7!$ ways.
If J&M sit on the long side they can be arranged in 20 ways, and the other guests in $7!$ Ways. Thus $20 \times 7!$

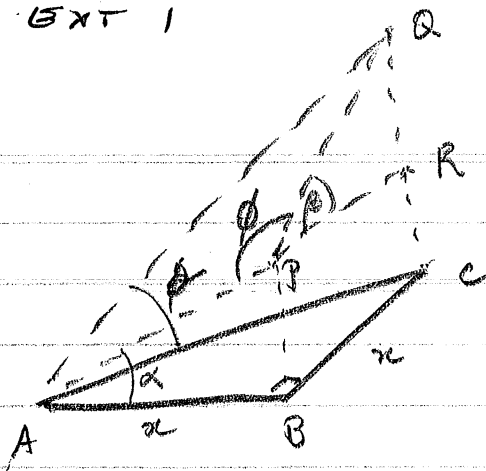
Hence there are $32 \times 7! = 161\,280$ ways.

- (ii) If John sits on the short side he has four seats available, and Mary (on the long side) has 5, thus $20 \times 7!$

But Mary may be the one on the short side.

Thus the total is $40 \times 7! = 201\,600$

Q7



$$(i) \quad \tan \alpha = \frac{BP}{x} = \frac{BP}{AB}$$

$$\therefore BP = x \tan \alpha$$

$$\tan \beta = \frac{QR}{x} = \frac{QR}{PR}$$

$$\therefore QR = x \tan \beta$$

$$QC = x \tan \alpha + x \tan \beta$$

$$= x (\tan \alpha + \tan \beta)$$

$$\tan \alpha = \frac{QC}{AC}$$

$$= \frac{x (\tan \alpha + \tan \beta)}{\sqrt{2} x}$$

$$= \frac{\tan \alpha + \tan \beta}{\sqrt{2}}$$

GIVEN

(2)

$$(ii) \quad \cos \phi = \frac{AP^2 + PQ^2 - AQ^2}{2 AP \cdot PQ}$$

$$= \frac{\frac{x^2}{\cos^2 \alpha} + \frac{x^2}{\cos^2 \beta} - \frac{2x^2}{\cos^2 \alpha}}{2 \cdot \frac{x}{\cos \alpha} \cdot \frac{x}{\cos \beta}}$$

$$= \frac{\sec^2 \alpha + \sec^2 \beta - 2 \left(1 + \frac{(\tan \alpha + \tan \beta)^2}{2} \right)}{2 \sec \alpha \sec \beta}$$

$$= \frac{\sec^2 \alpha + \sec^2 \beta - (2 + \tan^2 \alpha + 2 \tan \alpha \tan \beta + \tan^2 \beta)}{2 \sec \alpha \sec \beta}$$

$$= \frac{\cancel{\sec^2 \alpha} + \cancel{\sec^2 \beta} - \cancel{\sec^2 \alpha} - 2 \tan \alpha \tan \beta - \cancel{\sec^2 \beta}}{2 \sec \alpha \sec \beta}$$

$$= -\sin \alpha \sin \beta$$

GIVEN

(2)

$$\ddot{x} = 0$$

$$\dot{x} = 12 \cos 60^\circ$$

$$= 6$$

$$x = 6t$$

$$\ddot{y} = -10$$

$$\dot{y} = -10t + 12 \sin 60^\circ$$

$$= -10t + 6\sqrt{3}$$

$$y = -5t^2 + 6\sqrt{3}t$$

$$\therefore y = -5 \left(\frac{x}{6} \right)^2 + 6\sqrt{3} \times \frac{x}{6}$$

$$= -\frac{5x^2}{36} + \sqrt{3}x$$

(3)

$$\text{If } x = -y, -x = -\frac{5x^2}{36} + \sqrt{3}x$$

$$\therefore \frac{5x^2}{36} - (\sqrt{3}+1)x = 0$$

$$\therefore x(5x - 36(\sqrt{3}+1)) = 0$$

$$\therefore x = 0 \quad \text{or} \quad x = \frac{36(\sqrt{3}+1)}{5}$$

$$\therefore \text{GTF} = \frac{6(\sqrt{3}+1)}{5}$$

GIVEN

(3)

$$\dot{x} = 6$$

$$\dot{y} = -10 \times \frac{6(\sqrt{3}+1)}{5} + 6\sqrt{3}$$

$$= -12(\sqrt{3}+1) + 6\sqrt{3} = -6\sqrt{3} - 12$$

$$\text{Speed} = \sqrt{36 + (-6\sqrt{3} - 12)^2}$$

$$= \sqrt{36 + 108 + 144 + 144\sqrt{3}}$$

$$= \sqrt{288 + 144\sqrt{3}}$$

$$= 12\sqrt{2 + \sqrt{3}}$$

$$= 23.2$$

(2)